

22MA201-ENGINEERING MATHEMATICS II

Unit 1 &LP2- Method of Separation of Variables

2. Solution of First order and First Degree Differential Equations

2.1.Variables Separable Method

In solving differential equations, separation of variables was introduced initially by Leibniz and later it was formulated by John Bernoulli in the year 1694.

A first order differential equation is separable if it can be written as $h(y)y' = g(x)$ where the left side is a product of y' and a function of y and the right side is a function of x . Rewriting a separable differential equation in this form is called the method of separation of variables.

Finding a solution to a first order differential equation will be simple if the variables in the equation can be separated. An equation of the form $f_1(x)g_1(y)dx + f_2(x)g_2(y)dy = 0$ is called an equation with variable separable or simply a separable equation.

Rewrite the given differential equation as

$$\frac{f_1(x)}{f_2(x)}dx = -\frac{g_2(y)}{g_1(y)}dy \rightarrow (1)$$

Integration of both sides of (1) yields the general solution of the given differential equation as

$$\int \frac{f_1(x)}{f_2(x)}dx = -\int \frac{g_2(y)}{g_1(y)}dy + c, \text{ where } c \text{ is an arbitrary constant.}$$

Remarks

1. No need to add arbitrary constants on both sides as the two arbitrary constants are combined together as a single arbitrary constant.

2. A solution with this arbitrary constant is the general solution of the differential equation.

Problem 1:

$$\text{Solve } (1+x^2)\frac{dy}{dx} = 1+y^2.$$

Solution

$$\text{Given that } (1+x^2)\frac{dy}{dx} = 1+y^2. \quad \dots (1)$$

The given equation is written in the variables separable form

$$\frac{dy}{1+y^2} = \frac{dx}{1+x^2}. \quad \dots (2)$$

$$\text{Integrating both sides of (2), we get } \tan^{-1} y = \tan^{-1} x + C. \quad \dots (3)$$

$$\text{But } \tan^{-1} y - \tan^{-1} x = \tan^{-1} \left(\frac{y-x}{1+xy} \right). \quad \dots (4)$$

$$\text{Using (4) in (3) leads to } \tan^{-1} \left(\frac{y-x}{1+xy} \right) = C, \text{ which implies } \frac{y-x}{1+xy} = \tan C = a(\text{say}).$$

Thus, $y-x = a(1+xy)$ gives the required solution.

Problem 2:

Find the particular solution of $(1+x^3)dy - x^2 y dx = 0$ satisfying the condition $y(1) = 2$

Solution

$$\text{Given that } (1+x^3)dy - x^2 y dx = 0.$$

$$\text{The above equation is written as } \frac{dy}{y} - \frac{x^2}{1+x^3} dx = 0.$$

$$\text{Integrating both sides gives } \log y - \frac{1}{3} \log(1+x^3) = C_1, \text{ which implies,}$$

$$3 \log y - \log(1+x^3) = \log C.$$

$$\text{Thus, } 3 \log y = \log(1+x^3) + \log C,$$

$$\text{which reduces to } \log y^3 = \log C(1+x^3).$$

Hence, $y^3 = C(1 + x^3)$ gives the general solution of the given differential equation. It is given that when $x = 1$, $y = 2$. Then $2^3 = C(1+1) \Rightarrow C = 4$ and hence the particular solution is $y^3 = 4(1 + x^3)$.

First order separable ODEs

An ODE $\frac{dy}{dx} = F(x, y)$ is *separable* if we can write $F(x, y) = f(x)g(y)$ for some functions $f(x)$, $g(y)$.

Example

$$\frac{dy}{dx} = y \quad \text{IS separable,}$$

$$\frac{dy}{dx} = x^2 - y^2 \quad \text{IS NOT.}$$

Example 3. Find the general solution to the ODE

$$9y \frac{dy}{dx} + 4x = 0.$$

“Separating the variables”, we have

$$\begin{aligned} 9y dy &= -4x dx \iff \\ 9 \int y dy &= -4 \int x dx \\ \frac{9}{2} y^2 &= -\frac{4}{2} x^2 + C, \end{aligned}$$

i.e. the general solution is

$$\frac{x^2}{9} + \frac{y^2}{4} = K, \quad (K = C/36)$$

which describes a ‘family’ of ellipses.

We can check our solution by differentiating:

$$\frac{2}{9}x + \frac{2}{4}yy' = 0$$

i.e

$$9yy' + 4x = 0.$$

Example 4. Find the general solution to

$$\frac{dy}{dx} = \frac{y+1}{x+1}.$$

$$\begin{aligned}\Rightarrow \int \frac{1}{y+1} dy &= \int \frac{1}{x+1} dx \\ \Rightarrow \ln|y+1| &= \ln|x+1| + C.\end{aligned}$$

Use $\log\left(\frac{a}{b}\right) = \log a - \log b$:

$$\ln\left|\frac{y+1}{x+1}\right| = C,$$

or

$$\frac{y+1}{x+1} = e^C = K.$$

Again we can easily check this using differentiation.

Example 5. Solve the ODE

$$\frac{dy}{dx} = 1 + y^2$$

Separating variables:

$$\begin{aligned}\int \frac{dy}{1+y^2} &= \int dx \\ \Rightarrow \arctan y &= x + C \\ \Rightarrow y &= \tan(x + C).\end{aligned}$$

Once again, this is easily checked by differentiation.

Example 6. Solve

$$\frac{dy}{dx} - \frac{y(y+1)}{x(x-1)} = 0$$

finding y explicitly, i.e $y = f(x)$.

Solution: This equation is separable, thus separating the variables and integrating gives

$$\begin{aligned}\frac{dy}{dx} &= \frac{y(y+1)}{x(x-1)} \\ \int \frac{dy}{y(y+1)} &= \int \frac{dx}{x(x-1)}.\end{aligned}$$

To solve the integrals, use partial fractions:

$$\begin{aligned}\int \left[\frac{1}{y} - \frac{1}{y+1} \right] dy &= \int \left[-\frac{1}{x} + \frac{1}{x-1} \right] dx \\ \ln y - \ln(y+1) &= -\ln x + \ln(x-1) + C \\ \ln\left(\frac{y}{y+1}\right) &= \ln\left(\frac{x-1}{x}\right) + C \\ \frac{y+1}{y} &= e^{-C} \frac{x}{x-1}.\end{aligned}$$

Let $K = e^C$. Then

$$\begin{aligned}y &= (y+1) \left(\frac{x-1}{Kx} \right) \\ y \left[1 - \left(\frac{x-1}{Kx} \right) \right] &= \left(\frac{x-1}{Kx} \right) \\ y(Kx - x + 1) &= x - 1.\end{aligned}$$

$$\therefore y = \frac{x-1}{Kx-x+1}$$

is the explicit solution.

Example 7. Solve

$$(y+x^2y)\frac{dy}{dx}=1.$$

Solution:

$$\begin{aligned} y(1+x^2)\frac{dy}{dx} &= 1 \\ \int y \, dy &= \int \frac{dx}{x^2+1} \\ \frac{y^2}{2} &= \arctan x + C \end{aligned}$$

i.e. the solution is $y = \pm\sqrt{2\arctan x + 2C}$.

Discussion Questions:

1. What is the general idea behind the method of separation of variables in solving ODE?
2. Explain any limitations or restrictions associated with the method of separation of variables.
3. Can you provide an example of a differential equation where separation of variables cannot be applied?